

DS 200

Team 8 Project 1

April 5, 2024

Optimizing Volunteer Engagement at United Helpers Organization

Team Members

1. Linh Tran: Project Manager
2. Rachel Desulme: Analyst
3. Felicia Pranuthi Vijayarangam: Database Designer
4. Nathan Oyeniyi: Lead Developer
5. Otis Ely Murray: Developer
6. James Christian DiPasquale: Developer

A Description of the Process Adopted to Facilitate Collaborations/Sharing, Communications, Time-Management and Ensure Progress

To ensure our project's success through seamless collaboration, efficient time management, and clear communication, we've developed a comprehensive workflow process. We used SMS for swift meeting alerts, ensuring everyone is informed about upcoming gatherings in a timely manner. Microsoft Teams is our central hub for managing tasks, tracking progress, and storing documents, with the Kanban board providing a clear visual of responsibilities, deadlines, and task statuses to keep the team organized and aligned. We also use Teams for sharing images of queries and relevant documents, centralizing our data for easy access and review. To bolster our collaboration, we hold weekly in-person meetings every Tuesday and Thursday at the library, offering an opportunity for collective work and result assessments. On non-meeting days, team members progress independently but stay connected via Teams, with options for online participation or later updates for those unable to attend in person. Tasks are assigned with a 1 to 3-day completion window, promoting a consistent workflow. Moreover, we cultivate a supportive environment by encouraging those who finish early to aid others, ensuring collective efficiency and progress. This approach not only streamlines our operations but also fosters a cohesive and productive team dynamic.

Introduction and Background to the Data Requirements and End-User Data Needs

The United Helpers Organization provides aid to those affected by natural disasters. To ensure efficient operations, the organization needs records of their volunteers, tasks, and packing lists with their specific packages. Whether one's role is the overseer of the entire organization, a member of the logistics team, or a manager of a group under a specific task, this database created will improve coordination efforts throughout the entire relief process with real-time access to the information. Understanding end-user data needs is important for effectively serving these roles, therefore, the database includes a complete profile of the volunteers with their essential

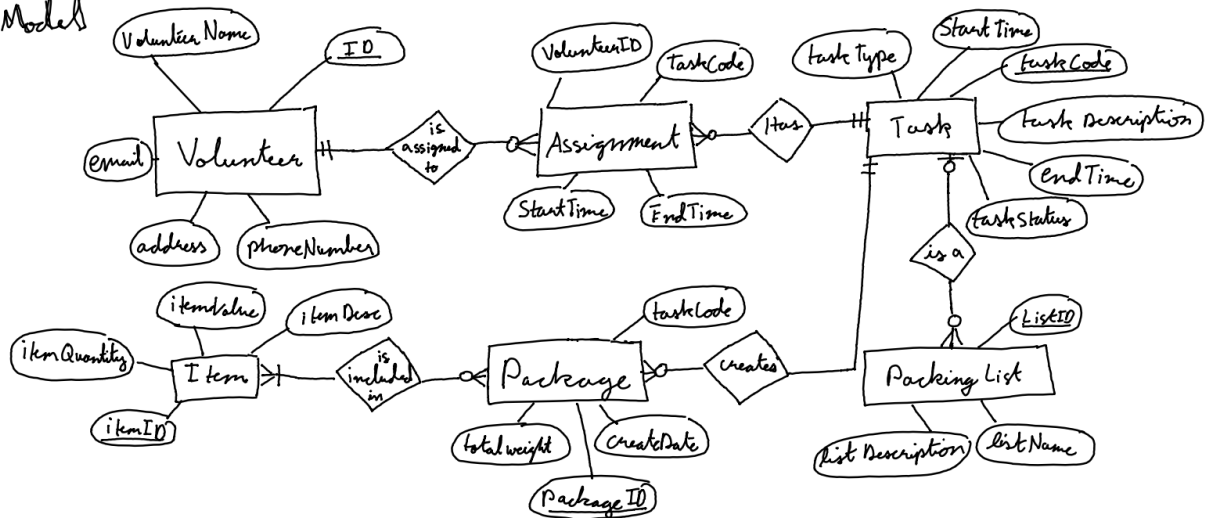
information such as address and contact details, a task management system, and a detailed packing list for the various types of relief packages. For further reporting, the database is also to have features established to monitor the status of each task. Implementing these needs can support the organization in its mission.

Detailed Clarification of Data Requirements in English

After reading the detailed description about the organization and assessing its needs, we came to develop what a successful database system that captures the information would look like as a team. There were six entities that stood out to us as necessary: Volunteers, Tasks, Assignments, PackingList, Package, and Items. In the Volunteers table, we included an ID for easy reference and tracking within the system. The name, address, phone number, and email are also included as the basic contact information. For the Task entity, the taskCode is the unique identifier and the attributes include taskDescription, taskType, taskStatus to record the details of each task, as well as the startTime and endTime to monitor volunteer activity and the duration. We established the associations between volunteers and tasks to track volunteer assignments with the Assignments entity, which allows multiple tasks per volunteer and multiple volunteers per task. The PackingList entity is used to define the packing lists which specify the contents of packages for different purposes, for example, medical and hygienic. It includes a unique ID identifier, along with the name and description to guide further preparation for the volunteers assigned for packing. The Package entity regards the individual packages made by the volunteers, also uniquely identified, including their weight and creation date. The Items entity contains the unique identifier, and records the details of quantity, description, and value (price). The entity would be useful in managing the inventory and tracking the contents of each package accurately. Implementing this database system with the necessary details will benefit the organization by enhancing its management systems and keeping the processes organized.

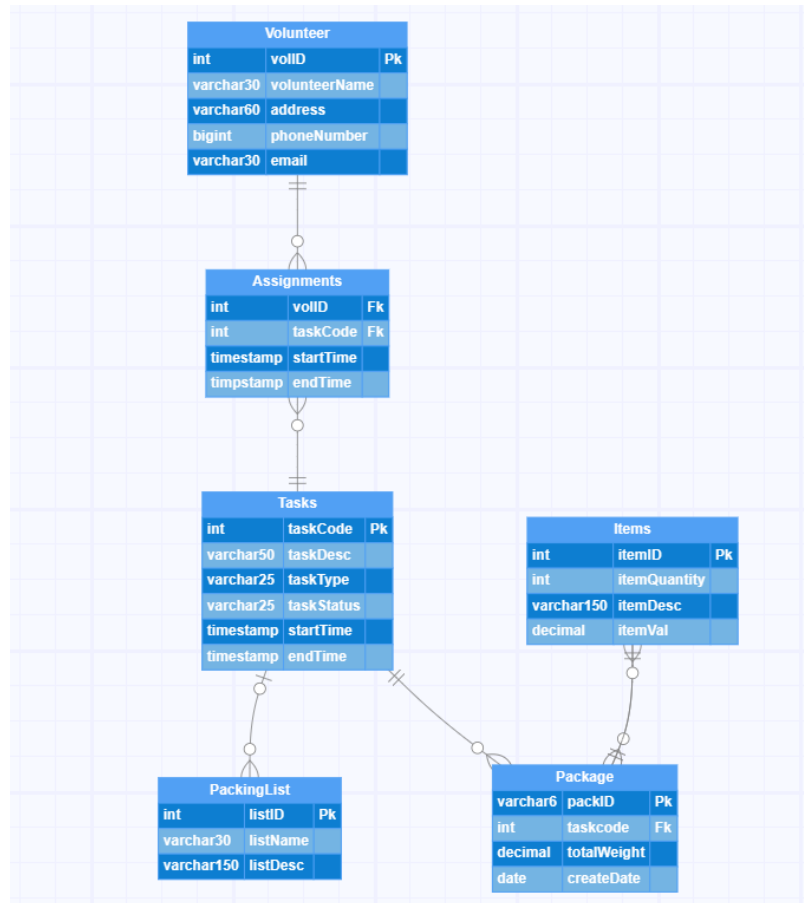
Conceptual Data Model

EER Model



One of the first steps in creating a database is creating a conceptual model. A conceptual model created by a database designer bridges the gap between the companies' requirements and the actual development of how these requirements will be implemented. It is not as specific as logical data model that developers use to understand what the variable values are. However, this model sketches out the entity, entity type, attribute, attribute types, relationship types, cardinality, and such. For our team's conceptual model, we came up with six entities: Volunteer, Assignment, Task, Item, Package, and Package List. For the Volunteer entity, our team came up with five attributes, name, ID, email, address, and phone number, to describe it. We underlined ID to indicate that its attribute type is a primary key. There is a one-to-many relationship between Volunteer and Assignment because a volunteer can be assigned to many assignments, even though an assignment can be assigned to one volunteer. An assignment entity has four attributes, VolunteerID, taskCode, startTime, and endTime. Each assignment created has a task. The relationship between an assignment and task is many to one because an assignment can have one task, but a task can have many assignments. The attributes for a task are taskType, startTime, taskCode, taskDescription, endTime, and taskStatus. Along with assignment, tasks forms a relationship between two other entities, package and packingList. A task is a packingList and has a one-to-many relationship because a task can have many packingList, but a packingList stems from a task. Notice that the cardinality also shows that a packingList is linked to either zero or one task. The attributes for a packingList are listID, listDescription, and listName. A Package has a one-to-many relationship from Task to Package. Task creates the entity Package, and the attributes of Package are taskCode, createDate, PackageID, and totalWeight. The package has a many to many relationships with Items. There are many items in each package and vice versa. Items have the attributes itemDesc, itemValue, itemQuantity, and itemID.

Logical Data Model



In our database design, we have crafted a logical ER diagram that illustrates the relationships between various entities. First, we begin with our volunteers, each possessing a unique identifier. These “Volunteers” can be assigned 0 or many tasks which is shown by our relation connection above. Observing the design, you'll notice the role of the "Assignments" entity, linking volunteers to their respective tasks. This entity, known as an Associative entity, serves as a central link for all tasks taken by one or more volunteers. By utilizing such an entity, we are enabled to lower the risk of data redundancy within our database. Moving along, we encounter the "Tasks" entity, which branches out to two distinct entities: "PackingList" and "Package." These entities come into play exclusively for packing-related tasks. In these scenarios, there exists the possibility of having zero or multiple packing lists and packages. However, it is important that each package is associated with just one packing task, while a packing list is linked to either zero or one task dedicated to packing. Lastly, we have the "Items" table. These individual items may find themselves enclosed within 0 or many packages, and packages can accommodate multiple items. Each entity maintains its unique identity through its designated ID, ensuring clarity and coherence within the database structure.

SQL Scripts for Our Database Tables and Implementation

Below are the create table statements for the entities ‘Volunteers’, ‘Tasks’, ‘Package’, ‘PackingList’, ‘Items’, and ‘Assignments’. Most of the primary keys and foreign keys have been set to ‘NOT NULL’ implying they must be existent, or they can’t be inserted. To record time in the tables below we used the data type ‘TIMESTAMP’ as it records both the date and hour-time at which things happened. Further, while most of the remaining attributes use common data types like varchar or int, we did have to use ‘bigint’ to record the phone numbers of the volunteers, because ‘mediumint’ was too small. We also used the ‘decimal’ data type so we could record things like price and weight more specifically. The code also contained the functions ‘PRIMARY KEY’, ‘FOREIGN KEY’, and ‘REFERENCES’ to show the direct relationships between the tables. Lastly, every table has a single primary key,

```

CREATE TABLE Volunteers (
  ID int NOT NULL,
  volunteerName varchar(30),
  address varchar(60),
  phoneNumber bigint,
  email varchar(30),
  PRIMARY KEY (ID)
);

CREATE TABLE Tasks (
  taskCode int,
  taskDescription varchar(50),
  taskType varchar(25),
  taskStatus varchar(25),
  startTime TIMESTAMP,
  endTime TIMESTAMP,
  PRIMARY KEY (taskCode)
);

CREATE TABLE Package (
  packageID varchar(6) NOT NULL,
  createDate DATE,
  totalWeight DECIMAL (4,2),
  taskCode INT NOT NULL,
  PRIMARY KEY (packageID),
  FOREIGN KEY (taskCode) REFERENCES Tasks(taskCode)
);

CREATE TABLE PackingList (
  ID varchar(5) NOT NULL,
  listName varchar(30),
  listDescription varchar(150),
  PRIMARY KEY (ID)
);

CREATE TABLE Items (
  itemID int NOT NULL,
  itemDescription varchar(150),
  itemValue decimal(4,2),
  itemQuantity int,
  PRIMARY KEY (itemID)
);

CREATE TABLE Assignments (
  VolunteerID int NOT NULL,
  TaskCode int NOT NULL,
  StartTime TIMESTAMP,
  EndTime TIMESTAMP,
  FOREIGN KEY (VolunteerID) REFERENCES Volunteers(ID),
  FOREIGN KEY (TaskCode) REFERENCES Tasks(taskCode),
  PRIMARY KEY (VolunteerID, TaskCode)
);

```

Below are the insert statements for the created tables above:

```

insert into Volunteers (ID, volunteerName, address, phoneNumber, email) VALUES
(4836, 'Joseph Murray', '659 Parkway Dr, Silverspring, MD 21202', 5782838293, 'jm463@gmail.com'),
(7483, 'Paul LaCosta', '364 Rocket Rd, Kalispell, MT 59901', 3748328477, 'pl2398@gmail.com'),
(5932, 'Jerome Smith', '493 House Av, Las Vegas, NV 88901', 1252834758, 'jsm928@gmail.com'),
(2333, 'James Lee', '298 Croft Rd, State College, PA 16801', 6502847293, 'jl2843@gmail.com'),
(1928, 'Garth Jenkins', '192 Northwoods Dr, New York City, NY 10001', 2938384612, 'gj2390@gmail.com')
;

insert into Tasks (taskCode, taskDescription, taskType, taskStatus, startTime, endTime) values
(1,'Answering the phone', 'Logistics', 'On Hold', '2024-04-01 08:00:00', '2024-04-01 16:00:00'),
(2,'Preparing packages', 'Packing', 'Occuring', '2024-04-02 09:30:00', '2024-04-01 16:30:00'),
(3,'Coordinate efforts', 'Reccuring', 'Occuring', '2024-04-03 10:00:00', '2024-04-03 17:30:00'),
(4,'Inventory management', 'Logistic', 'Completed', '2024-04-04 11:30:00', '2024-04-04 15:00:00'),
(5,'Delivering packages', 'Logistics', 'Completed', '2024-04-05 08:00:00', '2024-04-05 12:00:00'),
(6,'Collecting donations', 'Reccuring', 'Open', '2024-04-06 10:00:00', '2024-04-06 15:00:00')
;

insert into Package (packageID, createDate, totalWeight, taskcode) Values
('PKG001', '2024-04-04', 24.34,1),
('PKG002', '2024-04-05', 25.33,2),
('PKG003', '2024-04-06', 13.23,3),
('PKG004', '2024-04-07', 15.89,4),
('PKG005', '2024-04-08', 18.47,5),
('PKG006', '2024-04-09', 20.00,6)
;

insert into PackingList (ID, ListName, ListDescription) VALUES
('PL001', 'Childcare packages', 'Includes diapers, baby formula, and baby wipes'),
('PL002', 'Medical packages', 'Includes bandages, bottles of iodine, and bottles of hydrogen peroxide'),
('PL003', 'Clothing packages', 'Includes jackets, socks, and blankets for all ages'),
('PL004', 'Food packages', 'Includes canned goods, rice, and dried fruits'),
('PL005', 'Educational packages', 'Includes books, stationery, and educational toys'),
('PL006', 'Hygiene packages', 'Includes soap, toothbrushes, and toothpaste')
;

INSERT INTO Items (itemID, itemDescription, itemValue, itemQuantity) VALUES
(25,'Baby wipes', 3.00, 65),
(34,'Bandages', 4.39, 100),
(23,'Pairs of Socks', 6.28, 86),
(10,'Canned goods', 4.00, 200),
(19,'Book', 25.99, 25),
(56,'Soap', 5.49, 60)
;

```

CRUD Implementation for web access

Swagger
by SMARTBLAB

http://localhost:3000/Explore

standard public schema

12.0.2OAS 2.0

[Base URL: localhost:3000/]
http://localhost:3000/
PostgREST Documentation

Schemes

HTTP

Introspection

GET / OpenAPI description (this document)

country

GET /country

POST /country

DELETE /country

PATCH /country

Desktop 2

Breaking news
East Coast rocke...

Search

3:24 PM
4/5/2024

Containers [Give feedback](#)

Container CPU usage ⓘ

0.09% / 1200% (12 CPUs available)

Container memory usage ⓘ

119.7MB / 7.44GB

[Show charts](#)

Search

Only show running containers

	Name	Image	Status	CPU (%)	Port(s)	Last started	Actions
☐	compose-postgres-master		Running (4/4)	0.13%		22 hours ago	
☐	postgres-db ebe7a50acbd7	postgres:13-alpine	Running	0%	5432:5432	22 hours ago	
☐	postgres 2977fd301b3c	postgres/postgres:latest	Running	0.13%	3000:3000	22 hours ago	
☐	postgres-demo 081107ddd468	nginx:mainline-alpine	Running	0%	80:80	22 hours ago	
☐	swagger-ui						

Showing 8 items

DBBeaver 23.3.2 - Team Project

File Edit Navigate Search SQL Editor Data Tools

Database Navigator × Projects

Enter a part of object name here

> DBBeaver Sample Database (SQLite)

> postgres 2 - localhost:5432

▼ Databases

- > postgres
- > Team Project
 - > Schemas
 - > public
 - > Tables
 - > assignments
 - > items
 - > package
 - > packinglist
 - > tasks
 - > volunteers
 - > Foreign Tables
 - > Views
 - > Materialized Views
 - > Indexes
 - > Functions
 - > Sequences
 - > Data types
 - > Aggregate functions
 - > Event Triggers
 - > Extensions
 - > Storage

Project - General ×

Connection "postgres 2" configuration

Connection settings

PostgreSQL connection settings

▼ Connection settings

Initialization
Shell Command
Client identification
Transactions
General
Metadata
Errors and timeout
Data Transfer
Data Editor
SQL Editor

Main PostgreSQL Driver properties SSH SSL + Network configurations...

Server

Connect by: ☒ Host ☐ URL

URL:

Host:

Port:

Database:

Authentication

Authentication:

Username:

Password:

☒ Save password

Advanced

Session role:

Local Client:

[You can use variables in connection parameters.](#)

Driver name: PostgreSQL

[Driver Settings](#)

[Driver license](#)

Test Connection ...

OK

Cancel

Data Implementation Results

Below are the outputs of the created tables given the inserted values above as well as the queries and their resulting outputs.

1. Volunteers

volunteers Enter a SQL expression to filter results (use Ctrl+Space)					
	123 id	ABC volunteername	ABC address	123 phonenumber	ABC email
1	4,836	Joseph Murray	659 Parkway Dr, Silverspring, MD 21202	5,782,838,293	jm463@gmail.com
2	7,483	Paul LaCosta	364 Rocket Rd, Kalispell, MT 59901	3,748,328,477	pl2398@gmail.com
3	5,932	Jerome Smith	493 House Av, Las Vegas, NV 88901	1,252,834,758	jsm928@gmail.com
4	2,333	James Lee	298 Croft Rd, State College, PA 16801	6,502,847,293	jl2843@gmail.com
5	1,928	Garth Jenkins	192 Northwoods Dr, New York City, NY 10001	2,938,384,612	gj2390@gmail.com

2. Tasks

tasks Enter a SQL expression to filter results (use Ctrl+Space)						
	123 taskcode	ABC taskdescription	ABC tasktype	ABC taskstatus	starttime	endtime
1	1	Answering the phone	Logistics	On Hold	2024-04-01 08:00:00.000	2024-04-01 16:00:00.000
2	2	Preparing packages	Packing	Occuring	2024-04-02 09:30:00.000	2024-04-01 16:30:00.000
3	3	Coordinate efforts	Reccuring	Occuring	2024-04-03 10:00:00.000	2024-04-03 17:30:00.000
4	4	Inventory management	Logistic	Completed	2024-04-04 11:30:00.000	2024-04-04 15:00:00.000
5	5	Delivering packages	Logistics	Completed	2024-04-05 08:00:00.000	2024-04-05 12:00:00.000
6	6	Collecting donations	Reccuring	Open	2024-04-06 10:00:00.000	2024-04-06 15:00:00.000

3. Package

package Enter a SQL expression to filter results (use Ctrl+Space)				
	ABC packageid	createdate	123 totalweight	123 taskcode
1	PKG001	2024-04-04	24.34	1
2	PKG002	2024-04-05	25.33	2
3	PKG003	2024-04-06	13.23	3
4	PKG004	2024-04-07	15.89	4
5	PKG005	2024-04-08	18.47	5
6	PKG006	2024-04-09	20	6

4. Packing List

packinglist Enter a SQL expression to filter results (use Ctrl+Space)			
	ABC id	ABC listname	ABC listdescription
1	PL001	Childcare packages	Includes diapers, baby formula, and baby wipes
2	PL002	Medical packages	Includes bandages, bottles of iodine, and bottles of hydrogen peroxide
3	PL003	Clothing packages	Includes jackets, socks, and blankets for all ages
4	PL004	Food packages	Includes canned goods, rice, and dried fruits
5	PL005	Educational packages	Includes books, stationery, and educational toys
6	PL006	Hygiene packages	Includes soap, toothbrushes, and toothpaste

5. Items

items		Enter a SQL expression to filter results (use Ctrl+Space)			
Grid Text		123 itemid	ABC itemdescription	123 itemvalue	123 itemquantity
	1	25	Baby wipes	3	65
	2	34	Bandages	4.39	100
	3	23	Pairs of Socks	6.28	86
	4	10	Canned goods	4	200
	5	19	Book	25.99	25
	6	56	Soap	5.49	60

Query 1: This query lists all volunteers in the United Helpers Organization including their contact information.

Code:

select * from Volunteers;

Result: Includes the first and last name, address, phone number, and email of all the volunteers in the organization.

volunteers 1		Enter a SQL expression to filter results (use Ctrl+Space)				
select * from Volunteers						
Grid Text		123 id	ABC volunteername	ABC address	123 phonenumber	ABC email
	1	4,836	Joseph Murray	659 Parkway Dr, Silverspring, MI	5,782,838,293	jm463@gmail.com
	2	7,483	Paul LaCosta	364 Rocket Rd, Kalispell, MT 598	3,748,328,477	pl2398@gmail.com
	3	5,932	Jerome Smith	493 House Av, Las Vegas, NV 891	1,252,834,758	jsm928@gmail.com
	4	2,333	James Lee	298 Croft Rd, State College, PA	6,502,847,293	jl2843@gmail.com
	5	1,928	Garth Jenkins	192 Northwoods Dr, New York Ci	2,938,384,612	gj2390@gmail.com

Query 2: This query finds all volunteers who have not yet been assigned any tasks.

Code:

select * from Assignments where taskcode = null;

Result: The output is an empty table because all the volunteers have been assigned a task.

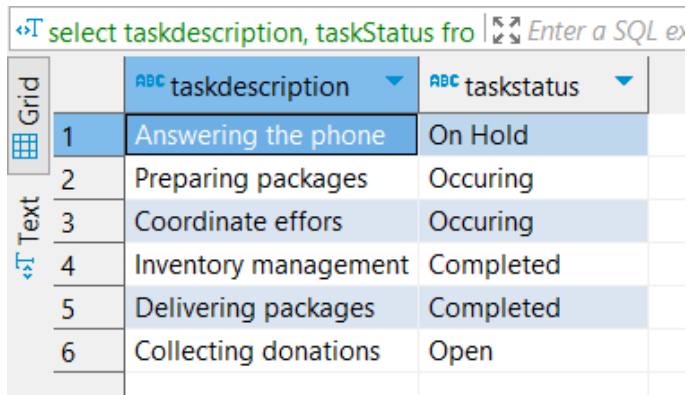
assignments 1		Enter a SQL expression to filter results (use Ctrl+Space)			
select * from Assignments w					
Grid		123 volunteerid	123 taskcode	starttime	endtime

Query 3: This query displays the status of each task.

Code:

```
select taskdescription, taskStatus from Tasks;
```

Result: The output is the name and status of each task.



The screenshot shows a database query editor with a SQL query entered: `select taskdescription, taskStatus from Tasks`. Below the query, a table grid displays the results. The grid has two columns: 'taskdescription' and 'taskStatus'. The results are as follows:

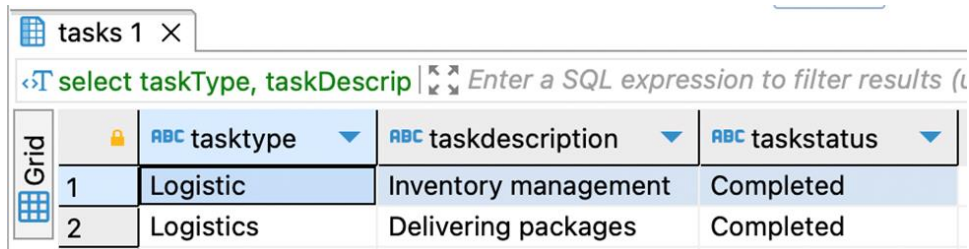
	taskdescription	taskStatus
1	Answering the phone	On Hold
2	Preparing packages	Occuring
3	Coordinate efforts	Occuring
4	Inventory management	Completed
5	Delivering packages	Completed
6	Collecting donations	Open

Query 4: This query lists all completed tasks.

Code:

```
select taskType, taskDescription, taskStatus from Tasks  
where taskStatus = 'Completed';
```

Result: The output is the task type along with their description, and the 'complete' status.



The screenshot shows a database query editor with a SQL query entered: `select taskType, taskDescription, taskStatus from Tasks where taskStatus = 'Completed'`. Below the query, a table grid displays the results. The grid has three columns: 'tasktype', 'taskdescription', and 'taskstatus'. The results are as follows:

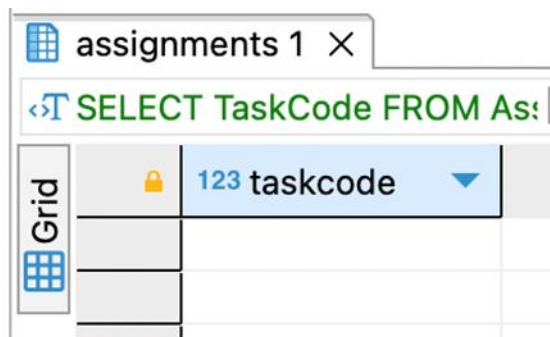
	tasktype	taskdescription	taskstatus
1	Logistic	Inventory management	Completed
2	Logistics	Delivering packages	Completed

Query 5: This query lists all tasks from the 'Assignments' table that have not been assigned any volunteers.

Code:

```
SELECT TaskCode FROM Assignments  
WHERE VolunteerID IS NULL;
```

Result: The result is an empty table because all tasks have been assigned to volunteers.



Query 6: This query lists all of the packages.

Code:

```
SELECT ID, listName, listDescription  
FROM PackingList;
```

Result: This table contains the ID, name, and description of each of the packages from the 'PackingList' table.

packinglist 1			
SELECT ID, listName, listDes			
Enter a SQL expression to filter results (use Ctrl+Space)			
Grid	ABC id	ABC listname	ABC listdescription
1	PL001	Childcare packages	Includes diapers, baby formula, and baby wipes
2	PL002	Medical packages	Includes bandages, bottles of iodine, and bottles of hydrogen peroxide
3	PL003	Clothing packages	Includes jackets, socks, and blankets for all ages
4	PL004	Food packages	Includes canned goods, rice, and dried fruits
5	PL005	Educational packages	Includes books, stationery, and educational toys
6	PL006	Hygiene packages	Includes soap, toothbrushes, and toothpaste

Query 7: This query selects the tasks that have been past due but have not been completed.

Code:

```
SELECT taskCode, taskDescription, startTime, endTime
FROM Tasks
WHERE endTime < NOW() AND taskStatus != 'Completed';
```

Result: This table shows the code, name, dates, and time when the incompleted tasks have been assigned.

tasks 1				
SELECT taskCode, taskDesc				
Enter a SQL expression to filter results (use Ctrl+Space)				
Grid	123 taskcode	ABC taskdescription	starttime	endtime
1	1	Answering the phone	2024-04-01 08:00:00.000	2024-04-01 16:00:00.000
2	2	Preparing packages	2024-04-02 09:30:00.000	2024-04-01 16:30:00.000

Documentation of Challenges

Linh Tran:

Lesson learned:

As a project manager, overseeing the development and implementation of our comprehensive workflow process has taught me valuable lessons in project management. One key lesson learned is the effectiveness of utilizing a Kanban board to assign tasks and balance team member schedules with project deadlines. The Kanban board serves as a visual representation of tasks, allowing us to prioritize and allocate responsibilities effectively. By utilizing this tool, we ensure that each team member understands their role and responsibilities while maintaining a balanced workload to meet project deadlines.

Challenges:

However, challenges inevitably arise, and one recurring challenge has been ensuring clear communication among team members, especially when some cannot attend in-person meetings. To address this challenge, I emphasize the importance of maintaining open lines of communication through various channels such as email, messaging platforms, or virtual meetings. Also, I make it a point to note absent members and follow up with them individually to ensure they are informed of their tasks and responsibilities. By proactively addressing communication gaps, we can minimize misunderstandings and keep the project on track.

Individual responsibilities:

Furthermore, balancing individual responsibilities within the team is essential for our collective success. As the project manager, I take on the responsibility of ensuring that each team member's workload is manageable and aligned with project objectives. This involves regular check-ins, progress updates, and adjustments to task assignments as needed. Additionally, I provide guidance and support to team members, addressing any challenges they may encounter along the way. By fostering a collaborative and supportive environment, we can effectively navigate obstacles and achieve our project goals together.

Rachel Desulme:

Lessons learned:

This is the first group project that I have been a part of that consistently met in person to collaborate on the work. In the past, most of the work was done individually and then grouped together at the end. I learned that meeting in person to work together helped us be on the same page with the quality of our work, and it was much easier for us to communicate and receive help from one another when we needed help with our tasks.

Challenges:

A challenge I had initially was understanding the context of the database description to flesh out the data requirements. Though there was creative freedom with the interpretation, I found it a bit difficult at first to align it with the specific needs of the project.

Individual responsibilities:

My responsibility as the Analyst was to flesh out the data requirements of the context for the organization's database, as well as framing the end-user queries and explaining the results. I also communicated with the database designers to make sure that the user queries would be successfully executed.

Felicia Pranuthi Vijayarangam: My role in this project was database designer. Therefore, my tasks were to sketch out our business requirements and pair them to what the developers must do. Before I started this project started, I struggled with understanding cardinality and relationship

types. As I read through the prompt and drew out how the relationships looked, I was able to understand if the entities had a one-to-one, many-to-many, or one-to-many relationship. In creating the conceptual model, I learned how to understand word problems and determine what entities, attributes, attribute types, and relationship types based on that. A major part of overcoming this challenge to ensure that our dev team had an accurate model to work from was drafting multiple EER diagrams and having the team approve. For example, when I drafted the EER diagram initially, there was debate whether there needed to be more entities. For example, we discussed whether to include a material entity as a go between tasks and package. Our initial EER model did not include this, but our second did, and we gave the Dev lead a preference whether to use it or not. When the database was being made, the Dev got back to us and said that there was no need for this connection because they felt it was better to include more attributes instead. One of the lessons I learned as a database designer is that it is important to get feedback from the Dev team to ensure that the model not only makes sense but is feasible. Our meeting twice a week helped us bring our ideas to the table and discuss the pros and cons of our design. Overall, my experience as a database designer, though challenging, taught me how to articulate my thoughts, sketch out an EER model, and work together with my team to ensure that the model was the increases efficiency.

Nathan Oyeniyi: Logical modeling and design is the blueprint for a successful running database. As the lead developer of our project's logical modeling and database design, I have learned the importance of these processes in ensuring the success and efficiency of our database implementation. Through attention to detail and effective collaboration with my teammates, I have learned to cycle through and understand the challenges associated with simplifying tables to fifth normal form, validating SQL syntax, establishing entity relations, and enforcing necessary constraints. This project provided the opportunity for us teammates to utilize our aggregated skill sets, received through academic lessons, towards real-world applications. Efficient communication with my team has been crucial in coming up with solutions to database-related challenges, using a changing environment which is necessary for our collective problem-solving. In my ability as the lead developer, I have created solid design framework to guide my teammates' efforts. Working closely with our project manager, I facilitated the creation of the initial Extended Entity-Relationship (EER) diagram, augmenting it with additional tables to preempt potential development hurdles. Subsequently, leveraging ER diagram tools, I was able to craft comprehensive diagrams exuding crucial components such as relations, keys, data types, and entity names. This approach ensured that our development team had a solid foundation upon which to create SQL queries and build the database backbone. Throughout the development process, I assumed the responsibility of analyzing SQL syntax to mitigate errors, thereby optimizing execution efficiency. Providing constructive feedback to developers, I facilitated iterative improvements, ensuring flawless execution and thorough performance. In essence, this project has not only improved my technical skills but also honed my leadership and collaboration skills, showing the importance of strong planning and effective communication in achieving group project objectives.

Otis Ely Murray: Over this project, I have learned that writing and finalizing code takes a lot of time. In our project, I was a developer and struggled with deciding which data types to represent the data in my tables. For example, whenever I tried to use predefined data types like 'DATETIME' and 'AUTO_INCREMENT', I would run into type errors even though those data types had previously worked on my device. I also struggled to ensure the foreign keys were placed correctly in the suitable tables, as I kept running into errors when I used the wrong keys. Further, some of my responsibilities during this project were to help write and edit the code to ensure that it worked and included everything in the project description. I helped to create 'insert' statements so that we could fill the tables with data. I also kept up with team communication, as I would send all my code through our group chat on Microsoft Teams. I also helped create and run the queries so that we could make sure the code was working as intended. Some of the lessons I learned during this project include but are not limited to team collaboration, delegation, organization, time management, and problem-solving skills. First, while team collaboration is seen as the job of the Project Manager, it is the role of every individual on the team. As the age-old saying goes, "There is no 'I' in the team," therefore, it was important for me to stay active in the team chat and through our other group chats, specifically SMS. Second, delegation and organization skills emerged as vital to our team's success. Learning to distribute tasks to others and ensuring everyone devoted time to the project allowed us to meet deadlines efficiently. Finally, the project allowed me to hone my problem-solving and time-management skills.

James Christian DiPasquale: Throughout the duration of this project, one of the primary hurdles I encountered revolved around the integration of the PostgREST API with our database infrastructure. This posed a significant challenge due to multiple versions of Postgres running concurrently. Consequently, when attempting to configure the new postgrest-db, I encountered persistent issues. In order to rectify this situation, I had to delve into the system files and meticulously remove the redundant versions of Postgres. This experience not only tested my technical skills but also provided valuable insights into problem-solving within database management. By working with the intricacies of setting up PostgREST API alongside our database, I gained a deeper understanding of the nuances involved in such configurations, ultimately enriching my skill set in database administration and API integration.

Conclusion

To summarize, our team has experienced great growth and achievement throughout this project. Linh Tran, the project manager, gained essential insights into efficient project management methods, notably the use of tools such as the Kanban board to balance team schedules and meet deadlines. Felicia Pranuthi Vijayarangam, the database designer, faced difficulty in

comprehending cardinality and relationship types while working with the development team to refine the model and assure its accuracy and viability. Rachel Desulme's experience demonstrated the value of in-person collaboration in improving job quality and communication efficiency. Nathan Oyenyi's leadership in database design demonstrated the need of precise preparation and effective communication in accomplishing project goals. Otis Ely Murray's coding problems highlighted the value of teamwork, delegation, and problem-solving skills in overcoming barriers. James Christian DiPasquale's experience with PostgREST API integration demonstrated the importance of flexibility and technical expertise when tackling complex tasks. Moving forward, our team intends to utilize these lessons by improving communication tactics, increasing collaboration, and creating a positive work atmosphere. We are confident and ready to take on new tasks by using our joint abilities and expertise.